

Chapter 2 – Software Development Models (Detailed Handbook)

1. Waterfall Model

Workflow Diagram



Phase-by-Phase Illustration

- Requirements: Gather and document business needs (BRD, SRS).
- Design: Create HLD, LLD, database and API designs.
- Development: Coding, code reviews, unit testing.
- Testing: Functional, integration, system and regression testing.
- Deployment: Release to production with validation.
- Maintenance: Bug fixes, enhancements and performance tuning.

Advantages

- Simple and easy to understand
- Well documented
- Suitable for fixed requirements
- Good for regulated industries

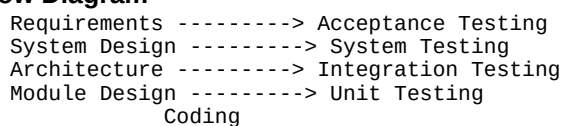
Disadvantages

- Difficult to accommodate changes
- Late customer feedback
- Requirement changes are costly

Real-world Example: Government tax filing system with fixed regulatory requirements.

2. V-Model

Workflow Diagram



Phase-by-Phase Illustration

- Each development phase has a corresponding testing activity.

- Testing planning starts before coding begins.
- Strong emphasis on verification and validation.

Advantages

- High quality
- Early defect detection
- Excellent traceability

Disadvantages

- Rigid process
- High documentation effort
- Not suitable for changing requirements

Real-world Example: Medical device software.

3. Incremental Model

Workflow Diagram

Requirements -> Increment 1 -> Release 1 -> Increment 2 -> Release 2 -> Final Product

Phase-by-Phase Illustration

- Develop features in multiple increments.
- Each increment delivers usable functionality.
- Customer feedback guides later increments.

Advantages

- Early business value
- Reduced project risk
- Simpler testing

Disadvantages

- Integration challenges
- Requires robust architecture

Real-world Example: Internet banking application.

4. Iterative Model

Workflow Diagram

Requirements -> Iteration 1 -> Feedback -> Iteration 2 -> Feedback -> Final Product

Phase-by-Phase Illustration

- Develop a basic version first.
- Refine through repeated iterations.
- Improve quality based on customer feedback.

Advantages

- Continuous improvement
- Flexible

- Supports evolving requirements

Disadvantages

- Scope creep
- More management effort

Real-world Example: E-commerce platform.

5. Spiral Model

Workflow Diagram

Planning -> Risk Analysis -> Engineering -> Customer Evaluation -> Repeat

Phase-by-Phase Illustration

- Plan objectives.
- Analyze risks.
- Develop prototype or product.
- Obtain customer feedback and repeat.

Advantages

- Excellent risk management
- Ideal for complex systems

Disadvantages

- Expensive
- Requires experienced project managers

Real-world Example: Air traffic control system.

6. RAD Model

Workflow Diagram

Business Modeling -> Data Modeling -> Process Modeling -> Application Generation -> Testing

Phase-by-Phase Illustration

- Build rapid prototypes.
- Validate with users frequently.
- Deliver quickly using reusable components.

Advantages

- Fast delivery
- High user involvement

Disadvantages

- Needs skilled developers
- Not ideal for very large projects

Real-world Example: HR leave management application.

7. Agile Model

Workflow Diagram

Product Backlog -> Sprint Planning -> Sprint -> Testing -> Review -> Retrospective -> Next Sprint

Phase-by-Phase Illustration

- Deliver software in short iterations.
- Collaborate continuously with customers.
- Adapt to changing requirements.

Advantages

- Highly flexible
- Frequent releases
- Continuous feedback

Disadvantages

- Requires engaged stakeholders
- Scope changes can affect planning

Real-world Example: Food delivery mobile application.

SDLC Model Selection Guide

Waterfall – Stable requirements

V-Model – Safety and compliance critical projects

Incremental – Early feature delivery

Iterative – Continuous refinement

Spiral – High-risk systems

RAD – Rapid business applications

Agile – Frequently changing requirements